# Exploratory Data Analysis (EDA)

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**Tools:**

Python (Pandas, Matplotlib, Seaborn)

**Objective:**

Extract insights from the Titanic dataset using visual and statistical exploration, identify relationships and trends between variables, and summarise key findings.

## 1. Dataset Overview

**The dataset contains 891 rows and 10 columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, and Fare.**

**Using .info(), it was found that most columns have no missing values, except Age, which has only 714 non-null values out of 891 — meaning 177 values are missing and will need to be handled in any further modeling.**

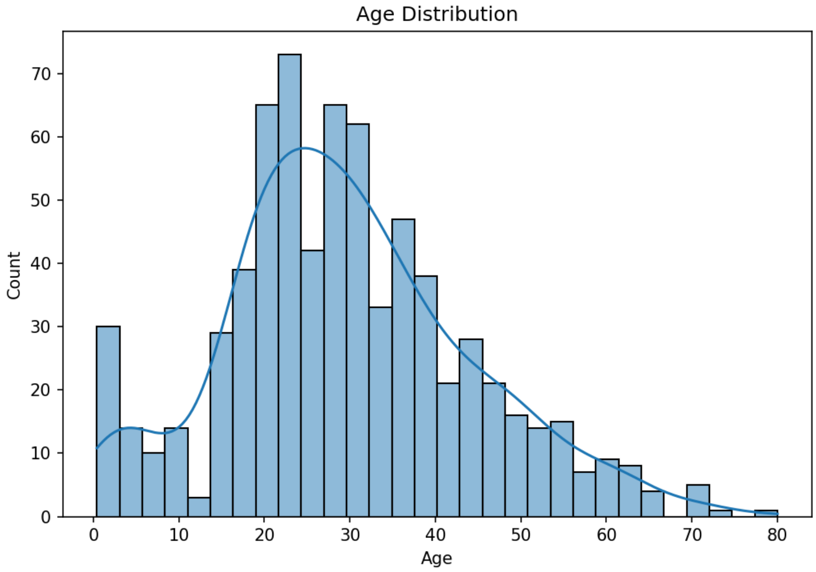
### Key statistics from .describe():

| Column | Mean | Min | Max |
|---|---|---|---|
| Survived | 0.38 | 0 | 1 |
| Age | 29.7 | 0.42 | 80 |
| Fare | 32.2 | 0.0 | 512.3 |
| Pclass | 2.3 | 1 | 3 |

### Category counts (.value_counts()):

- Sex — Male: 577, Female: 314
- Survived — Died (0): 549, Survived (1): 342 (~38% survival rate)
- Pclass — Class 3: 491, Class 1: 216, Class 2: 184

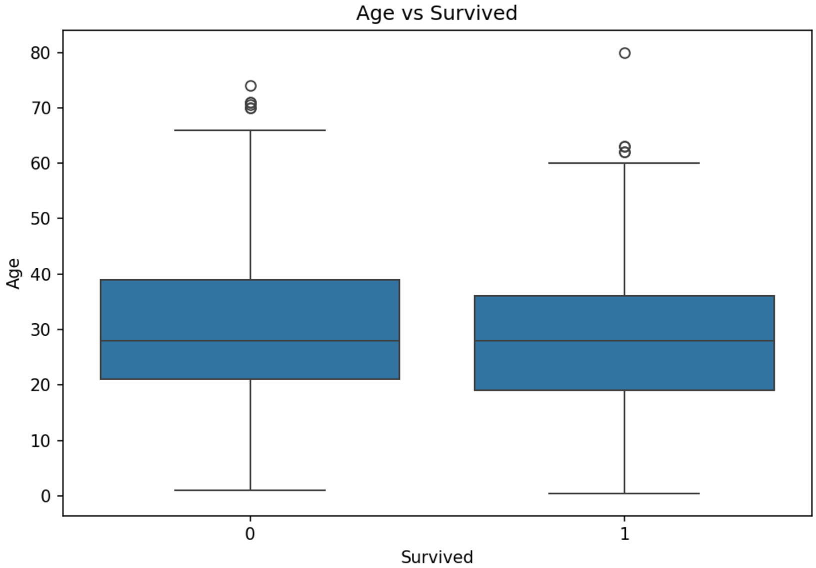
## 2. Age Distribution (Histogram)




**Observation:**

**The majority of passengers fall between 20 and 35 years, forming the tallest cluster of bars in the chart. A distinct secondary spike appears near age 0-5, indicating a meaningful number of infants and young children were on board, likely traveling with family. The overall shape is right-skewed, with a long thinning tail stretching toward 60-80 years, showing that elderly passengers were comparatively rare. This spread suggests the ship's population was dominated by young adults, which is typical of early 1900s transatlantic travel and immigration patterns.**

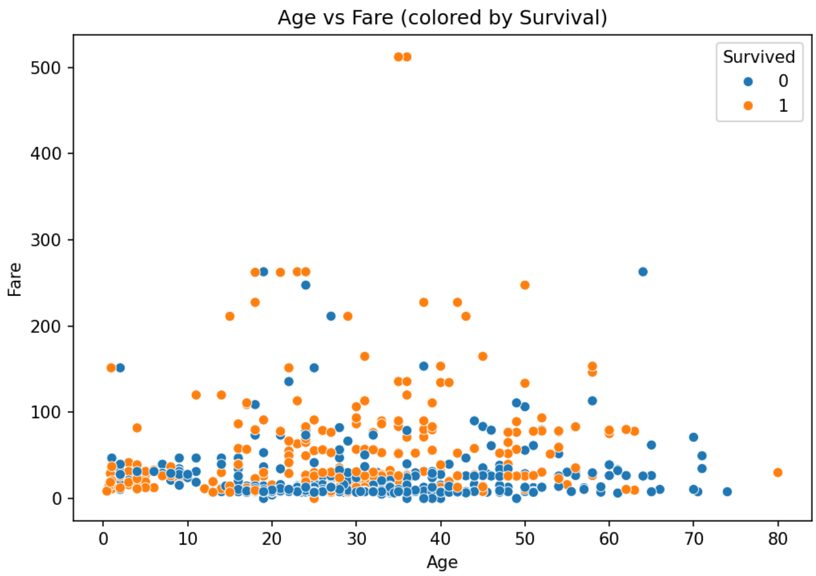
## 3. Age vs Survived (Boxplot)




**Observation:**

**Both groups — survivors and non-survivors — share a very similar median age of roughly 28 years, and their interquartile ranges overlap heavily, which indicates that age by itself was not a decisive factor in who survived. A handful of outliers appear above 65 years in both categories, showing a small number of elderly passengers on both sides of the outcome. This reinforces that survival was driven more by other factors such as class, fare, and sex rather than age alone, and that age should be examined together with these variables rather than in isolation.**

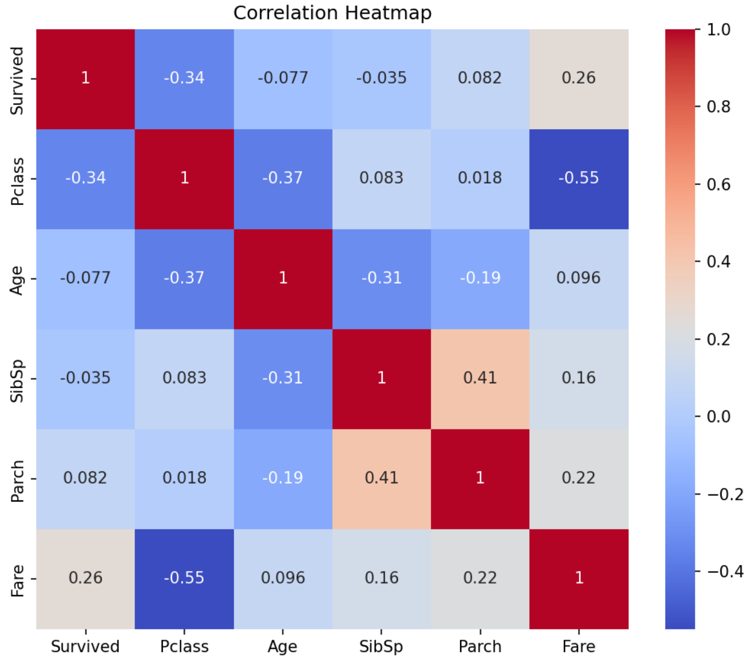
## 4. Age vs Fare (Scatterplot, colored by Survival)




**Observation:**

**Passengers who paid noticeably higher fares, roughly above 100, are overwhelmingly marked orange (survived), while the dense cluster of low-fare passengers below 50 skews strongly blue (did not survive). This pattern holds fairly consistently across all age groups, suggesting fare — and by extension the wealth or class it represents — had a real influence on survival outcome, independent of how old the passenger was. It also hints that higher-fare cabins may have been positioned closer to lifeboats or received faster evacuation priority during the disaster.**

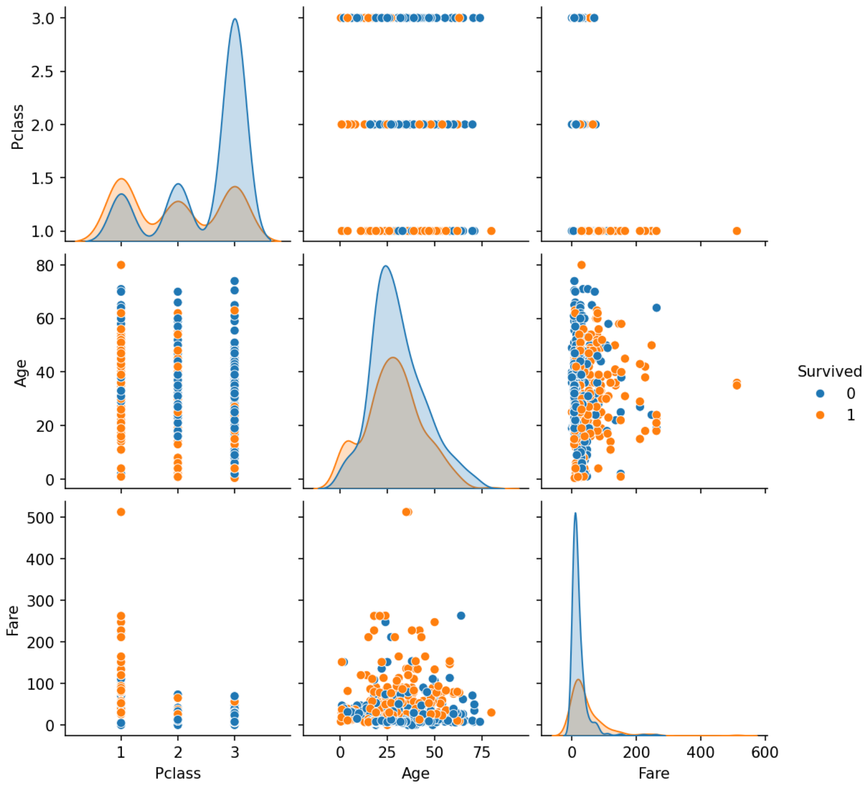
## 5. Correlation Heatmap




**Observation:**

**Fare shows the strongest positive correlation with Survived at 0.26, confirming that wealthier passengers had a somewhat better chance of survival. Pclass correlates negatively with Survived at -0.34, meaning passengers in 1st class (numerically lower Pclass) survived more often than those in 3rd class. As expected, Pclass and Fare are strongly negatively correlated at -0.55, since higher-class tickets cost more. SibSp and Parch show a moderate positive correlation of 0.41, which makes sense as both describe family size aboard. Age shows weak correlation with every other variable, reinforcing that it is not a strong standalone predictor of survival.**

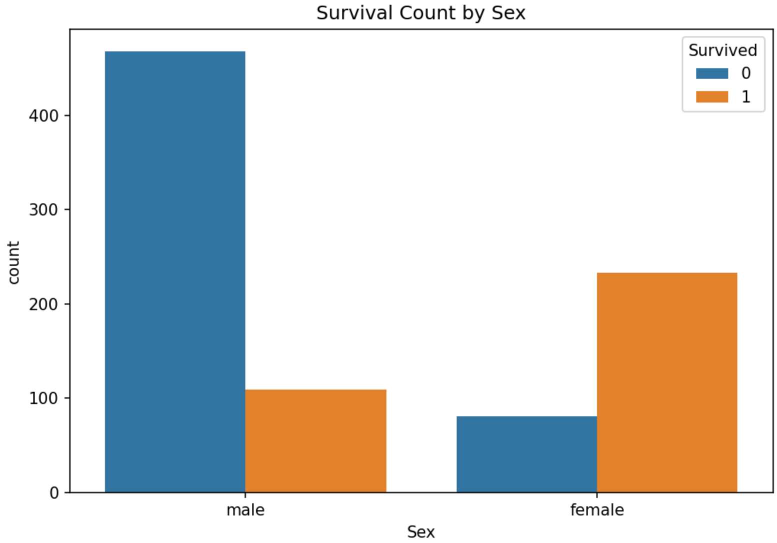
## 6. Pairplot (Survived, Pclass, Age, Fare)




**Observation:**

**The Pclass panel shows survivors (orange) more evenly spread across classes 1 and 2, while non-survivors (blue) are heavily concentrated in class 3, visually confirming the class-survival link seen earlier. The Fare panel shows survivors' distribution stretching further into higher fare values, again tying fare to better survival odds. The Age panels show largely overlapping distributions for both outcomes, with no sharp separation, confirming age plays a secondary role compared to class and fare in determining who survived.**

## 7. Survival Count by Sex




**Observation:**

**This is the clearest pattern in the entire dataset: the vast majority of male passengers, roughly 470, did not survive, while only about 110 did. Female passengers show the exact opposite trend — around 230 survived compared to only about 80 who did not. This sharp contrast reflects the historically documented "women and children first" evacuation protocol followed during the disaster, and statistically makes sex the single strongest predictor of survival among all variables examined in this analysis, outweighing even fare and class individually.**

# 8. Summary of Findings

- 1. About 38% of passengers survived overall.
- 2. females had a much higher survival rate than males.
- 3. Passenger class (Pclass) mattered — 1st class passengers survived more than 3rd class.
- 4. Fare correlated with survival — higher fare passengers survived more, tying back to class.
- 5. Age alone did not show a strong pattern, though very young children had slightly better survival odds.
- 6. Age has 177 missing values that would need handling in further analysis.

**Outcome:**

This task helped in developing skill in identifying patterns, trends, and anomalies within a real-world dataset using univariate, bivariate, and multivariate visual analysis techniques.